

Philosophers Project - Low Level OS, Threads and Concurrency Roadmap

Phase 1: CPU Fundamentals - Registers (RIP, RSP, general registers) - Fetch, Decode, Execute cycle - Function calls and stack frames - Process memory layout (Text, Data, Heap, Stack) - Recursion, malloc, memory regions Phase 2: Processes - What a process is - PID, memory, files, CPU state - task_struct concept - fork(), execve(), wait() - ps, top, htop Phase 3: Why Threads Exist - Single execution flow vs multiple execution flows - Shared resources: code, heap, globals, files - Private resources: stack, registers, instruction pointer Phase 4: What a Thread Really Is - Registers - Instruction Pointer - Stack - Kernel scheduling Phase 5: Context Switching - Saving/restoring registers - Timer interrupts - Switching execution between threads Phase 6: Scheduling - Running, Ready, Blocked states - Time slices - Preemption - Round Robin - Linux CFS Phase 7: Race Conditions - Read-modify-write operations - Why counter++ is not atomic - Data corruption examples Phase 8: Mutexes - Mutual Exclusion - Critical sections - pthread_mutex_lock() - pthread_mutex_unlock() Phase 9: Mutex Internals - Atomic operations - Compare-And-Swap (CAS) - Futex - Fast path vs slow path Phase 10: Memory Visibility - CPU caches - Cache coherence - Memory ordering - Memory barriers Phase 11: Condition Variables - Busy waiting vs sleeping - pthread_cond_wait() - pthread_cond_signal() - pthread_cond_broadcast() Phase 12: Why Condition Variables Need Mutexes - Lost wakeup problem - Atomic check-and-sleep - Correct waiting patterns Phase 13: Deadlocks - Mutual exclusion - Hold and wait - No preemption - Circular wait - Dining Philosophers deadlock Phase 14: Linux Internals - Processes - Threads - Scheduler - Futexes - Virtual memory Recommended Reading: - Operating Systems: Three Easy Pieces (OSTEP) - Modern Operating Systems - Linux Kernel Development Phase 15: After Philo - CPU architecture - Operating systems - Linux scheduler - Memory models - Atomics - Lock-free programming - Linux kernel source code